Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
PI Name:	Baldwin, Kenneth M. Ph.D.		
Project Title:	Role of Muscle Loading on Mechanisms of Protein Translation and the Impact on Unloading-Induced Atrophy		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kmbaldwi@uci.edu	**Fax:**	FY 949-824-8540
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-7192
Organization Name:	University of California, Irvine		
PI Address 1:	Medical Sciences I, Room D340		
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PI Web Page:	http://muscle.biomol.uci.edu/		
City:	Irvine	**State:**	CA
Zip Code:	92697-4560	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	02/29/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	15
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	N/A		
Key Personnel Changes/Previous PI:	Baldwin, Kenneth		
COI Name (Institution):	Adams, Gregory (University of California, Irvine) Caiozzo, Vincent (University of California, Irvine) Haddad, Fadia (University of California, Irvine)		
Grant/Contract No.:	NCC 9-58-MA00202		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In this project our goals were two- fold: 1) ascertain a better understanding of the cellular factors that cause muscle atrophy; 2) develop a resistance training protocol that can successfully create an anabolic state in the muscle that could serve to counteract the catabolic state that is associated with models of muscle unloading. Our findings suggest that in models cause rapid muscle atrophy there are fundamental reductions in the ability of key skeletal muscle genes to transcribe the message and the cell machinery to translate the message into protein. This deficit occurs in the face of elevations in the protein degradation systems that cause the normal breakdown in these proteins. Such an imbalance results in net protein loss in the muscle, thereby cause the muscle cell to become smaller. Resistance training improves the status of the muscle by slowing down the protein breakdown process while at the same time enhancing the muscle to synthesize new contractile protein. We are confident that by selecting the right prescription of resistance training we will be successful in preventing the muscle wasting that occurs in space travelers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The problem of sarcopenia or muscle wasting associated with aging affects every human being over 40 years of age. Our research on the mechanisms of muscle atrophy and the use of resistance exercise, in the face of muscle unloading, will provide practical knowledge and useful strategies for ameliorating or slowing this critical health problem.
Task Progress:	The project for the last year was complete and resulted in a publication in the Journal of Applied Physiology
Bibliography Type:	Description: (Last Updated: 04/21/2011)
Articles in Peer-reviewed Journals	Adams GR, Cheng DC, Haddad F, Baldwin KM. "Skeletal muscle hypertrophy in response to isometric, lengthening, and shortening training bouts of equivalent duration. " J Appl Physiol. 2004 May;96(5):1613-8. PMID: 15075307 , May-2004
Articles in Peer-reviewed Journals	Haddad F, Baldwin KM, Tesch PA. "Pretranslational markers of contractile protein expression in human skeletal muscle: effects of limb unloading plus resistance exercise. " J Appl Physiol. 2005 Jan;98(1):46-52. Epub 2004 Aug 6. PMID: 15298986 , Jan-2005
Articles in Peer-reviewed Journals	Haddad F, Roy RR, Zhong H, Edgerton VR, Baldwin KM. "Atrophy responses to muscle inactivity: I. Cellular markers of protein deficits. " J Appl Physiol. 2003 Aug;95(2):781-90. PMID: 12716870 , Aug-2003
Articles in Peer-reviewed Journals	Haddad F, Roy RR, Zhong H, Edgerton VR, Baldwin KM. "Atrophy responses to muscle inactivity: II. Molecular markers of protein deficits. " J Appl Physiol. 2003 Aug;95(2):791-802. PMID: 12716877 , Aug-2003